United International University (UIU)

Dept. of Computer Science & Engineering (CSE)

Exam Name: Final Term Exam   Trimester: Spring 2026
Course Code: PMG 4101, Course Title: Project Management
Total Marks: **40**   Duration: 2 hours

Any examinee found adopting unfair means will be expelled from the trimester / program as per UIU disciplinary rules.

Answer all the questions.

Consider the following Scenario:

In Bangladesh, land offices still depend a lot on paper files, manual registers, and disconnected systems. Because of this, people often have to wait a long time for services like land registration, mutation, ownership checks, or tax updates. Documents can be hard to find, records don't always match across offices, and the lack of a single system increases mistakes and the risk of fraud. To fix these issues, the Ministry of Land has started building a National Digital Land Record and Property Management System, and your company has been chosen to design and implement it across the country.

The system will be used by citizens, land officers, surveyors, tax officials, legal teams, and ministry staff, and it needs to handle a large number of users at the same time, especially during busy periods. The project team includes 18 senior engineers working on architecture, security, GIS, and infrastructure, and 32 junior staff handling UI development, testing, training, support, and digitizing old records. Regular updates, demos, and review meetings with government stakeholders are expected throughout the project.

As the project manager, you are leading the development of a web-based system with role-based access. Citizens will be able to search land records, apply for mutation, pay fees online, track their applications, and download approved documents. Land officers will review and process applications through defined workflows, while surveyors will update land maps and property details digitally. The system will bring everything into one place—land ownership records, maps, tax details, legal documents, and dispute histories. It will support approvals, digital signatures, notifications, deadlines, audit logs, and reporting, and will connect with national ID verification, payment systems, GIS services, and government treasury systems. Since land data is highly sensitive, strong security like encryption, access control, backups, disaster recovery, version control, and audit tracking is essential.

The UI part is fairly simple, but the real challenge is behind the scenes—handling large amounts of data, complex database queries, and multiple external integrations. Overall, complexity is high for F1, F2, F3, F6, and F7; moderate for F4 and F5; essential for F8, F9, F10; and average for the rest. Based on this, the estimated productivity is **110 Lines of Code per Function Point (LOC/FP)**.

1. Consider the above scenario to calculate-

i. The ***CT (count total)*** and ***Value Adjustment Factor (VAF)*** for this project.

ii. Calculate ***FP (Function point)*** of this project using the above VAF.

[CO3]   3+2+1 = 6

2. i. Explain what type of project the above scenario represents. Is it ***Organic***, ***Semi-Detached*** or ***Embedded***? – Give proper reasoning.

ii. Find all the ***COCOMO estimating parameters*** for the project mentioned in the above scenario [show all calculations i.e., efforts, duration, productivity, budget etc].

[CO3] 2+4 = 6

3. i. Define ***Change Control*** concept and Explain the role of ***Developer*** in ***Change control board (CCB)***.

ii. Discuss the ***steps*** (basic course of events) and ***alternate steps*** for Change Control that should be followed in a software project management process.

[CO4] 2+4 = 6

4. Develop a ***Gantt-chart*** for the above project scenario by identifying the ***tasks*** and ***tasks' dependency***. Also, mention different assumptions or considerations that need to be considered (e.g., weekly holidays and government holidays) for making a pragmatic estimation. **[CO5] 4+2 = 6**

5. Which ***Review method*** will be appropriate for reviewing the project above, mention your justification. Also describe the ***Basic Course of Events*** of that review process.

[CO4] 2+4 = 6

6. List the differences between ***Mature & Immature Organization*** based on CMMI model/concepts. Describe the **Eight Core Principles of the ISO 9000.**

[CO5] 5

7. Progress report submission and presentation on estimation through COCOMO model and Function point methods for a group project has been evaluated earlier also show the project management activities through the suitable PM Tool. (**Do not answer this question**, **already completed in classroom**). **[CO5] 5**

Parameter	**Simple**	**Avg**	**Complex**
EI	3	4	6
EO	4	5	7
EQ	3	4	6
ILF	7	10	15
EIF	5	7	10

Project Type	**a**	**b**	**c**	**d**
Organic	2.4	1.05	2.5	0.38
Semidetached	3.0	1.12	2.5	0.35
Embedded	3.6	1.20	2.5	0.30